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**PAPER**

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# Classification of Motor Imagery EEG signals using high resolution time-frequency representations and convolutional neural network

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Keywords: Brain computer interface (BCI), Complex Morlet Wavelet (CMW), EEG, Motor Imagery (MI), Time-frequency representations (TFR), CNN

## Abstract

A Motor Imagery (MI) based Brain Computer Interface (BCI) system aims to provide neuro-rehabilitation for the motor disabled people and patients with brain injuries (e.g., stroke patients) etc. The aim of this work is to classify the left and right hand MI tasks by utilizing the occurrence of event related desynchronization and synchronization (ERD\ERS) in the Electroencephalogram (EEG) during these tasks. This study proposes to use a set of Complex Morlet Wavelets (CMW) having frequency dependent widths to generate high-resolution time-frequency representations (TFR) of the MI EEG signals present in the channels C3 and C4. A novel method for the selection of the value of number of cycles relative to the center frequency of the CMW is studied here for extracting the MI task features. The generated TFRs are given as input to a Convolutional neural network (CNN) for classifying them into left or right hand MI tasks. The proposed framework attains a classification accuracy of 82.2% on the BCI Competition IV dataset 2a, showing that the TFRs generated in this work give a higher classification accuracy than the baseline methods and other existing algorithms.

## 1. Introduction

Brain Computer Interface (BCI) is a communication control system through which an individual can send commands or messages to the external world without relying on the brain's normal neuromuscular pathways [1]. Such a system can be used to identify the intent of people who have severe neuromuscular disorders, like amyotrophic lateral sclerosis, brainstem stroke, spinal cord injuries, etc. by processing the electrophysiological signals collected from the brain. There are several types of electroencephalogram (EEG) based BCI systems depending on different modalities of EEG like P300 event related potentials, Sensory Motor Rhythms (SMR), Slow Cortical Potentials (SCPs) and Steady State Visually Evoked Potentials (SSVEP) [2]. SMRs are known to be movement-modulated rhythms that are recorded over the sensorimotor cortex. The SMRs are the brain's oscillations that fall in the frequency range of (8–30) Hz, where (8–13 Hz) band is known as the mu rhythm and (18–30 Hz) band is known as the beta rhythm [2]. Motor Imagery is a cognitive process of imagining the movement of a body part without moving it [3]. When a person performs or imagines a unilateral limb movement, the amplitude of the SMRs in the contralateral cortex of the brain attenuates. This is called the event related desynchronization (ERD). Conversely, immediately after the imagery movement, an increase in the SMR, called as event related synchronization (ERS), occurs in the contralateral region [4]. The ERD/ERS phenomenon of left and right hand motor imagery mainly occurs in the C3 and C4 regions of the cerebral cortex. In a BCI system, the recognition and interpretation of the MI EEG signal is difficult due to its low signal to noise ratio, non-linearity, and non-stationary nature [5, 6]. It also exhibits trial-to-trial and subject-to-subject variability. Hence feature extraction and classification methods are to be used in order to increase the accuracy of classification of the MI tasks.

The most commonly used feature extraction techniques for MI EEG signals include Autoregressive (AR) model [7, 8], Common Spatial Patterns (CSP) [9–12], CSP variants such as Filter-Bank CSP (FBCSP) [13, 14], Short Time Fourier Transform (STFT),

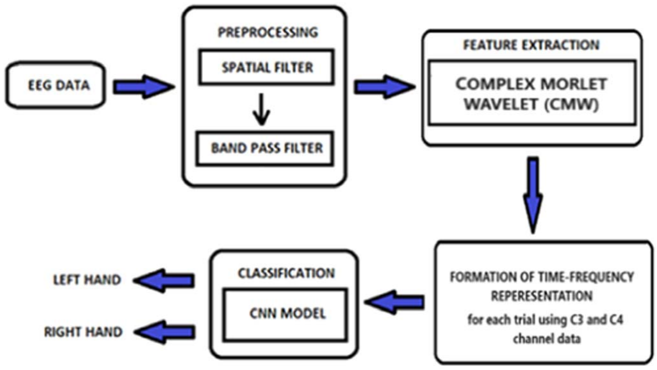


**Figure 1.** Framework of the MI-EEG Classification system.

Wavelet Transform (WT) and so on. As the brain's oscillations are transient, the EEG signal needs to be localized in both time and frequency in order to evaluate them [15]. The most widespread approaches used to generate the time-frequency representations (TFR) are Short Time Fourier Transform [16–18], Wavelet Transform [19–21] etc. The major drawback of a STFT is that it uses a window of fixed length, which decides the tradeoff between the time and frequency information. When the size of the window is shortened to achieve a good temporal resolution, it causes a degradation in the frequency resolution [22]. In order to compensate for these shortcomings, the WT is used to perform a multiresolution analysis of the MI EEG signals, which means that it analyzes signals at different frequencies with varying resolutions. The WT has a relatively good temporal resolution throughout but undergoes degradation in frequency resolution with increasing frequency [23]. Therefore, STFT and WT provide a suboptimal tradeoff between time and frequency resolutions which hinders the analysis of the neuronal signals.

This study proposes to generate a sharper TFR of the EEG signals by using a set of wavelets whose widths are taken as a function of the frequencies which are to be analysed. The arithmetic mean of their responses are used to in order to gain an optimal temporal and frequency resolution. The traditional classifiers that are being used for the classification of the MI-EEG signal features are Linear Discriminant Analysis (LDA) [24], Support Vector Machines (SVM) [25], k-Nearest Neighbor (KNN) [26] etc.

Deep learning methods are recently being used in BCI studies as they effectively deal with non-stationary data and can automatically learn the underlying features from the data [27–29]. In [30] different architectures of the convolutional neural networks (CNN) named ConvNets were built using the recent advances in deep learning, for decoding the MI tasks. Although they performed as good as FBCSP, they had many hyperparameters to learn and required a large amount of training data to have a better performance. Multiscale CNN models with different convolution kernel sizes are built in [31, 32], to extract effective features from the MI data and enhance the performance of the classifier. In [31], adaptive transfer learning based CNN model is used, thereby attaining a higher classification accuracy than most of the existing methods by using the MI data from 22 channels. The method used in [32] gives better performance than the state-of-the-art methods using data augmentation techniques on a smaller MI dataset. Training a CNN model for the smaller BCI MI datasets and achieving a better performance is a challenging task. In this work a Convolutional Neural Network (CNN) with two convolutional layers is built to classify the TFR of the left or right hand MI EEG signals. This framework gives a better classification performance on BCI Competition IV dataset 2a with respect to the other existing works by using the data from lesser number of channels. Hence it shows that the TFR generated in this work can extract the time-frequency (TF) features from the MI EEG signals with a higher resolution thereby giving a better classification performance.

The major contributions of this work are listed below:

(a) High resolution time-frequency features of the MI EEG signals are extracted using a set of wavelets whose width is a function of the frequency.

(b) A CNN model with two convolutional layers is proposed for classifying the TFR of the left or right hand MI EEG signals using two channels C3 and C4.

## 2. Materials and methods

### 2.1. The system framework

The figure 1 given below depicts the framework of the proposed BCI system for classifying the MI-EEG signal into two classes, left or right hand movement. The MI-EEG data show significant decrease in the amplitude of the mu (8–13 Hz) and beta (18 to 30 Hz) frequency bands over the contralateral sensory motor area.

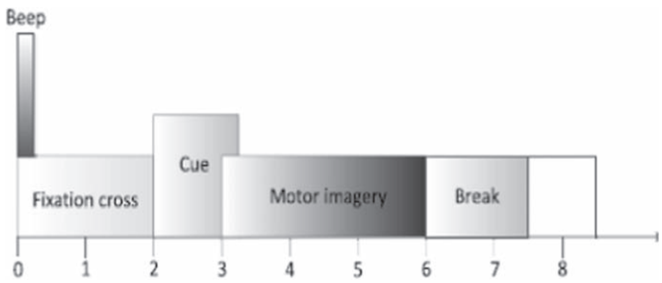


**Figure 2.** Experimental paradigm for BCI Competition IV dataset 2a [33].

Hence for each subject, the Electroencephalogram (EEG) data recorded for each trial over C3 and C4 electrodes are isolated from the dataset for further processing. As shown in figure 1, the proposed method consists of four steps including (1) Preprocessing (2) Feature Extraction (3) Formation of TFR (4) Classification. In the following sections, the different parts of the proposed BCI system will be explained in detail.

### 2.2. Dataset

The current study uses the following public domain MI EEG dataset to evaluate the performance of the proposed system.

*2.2.1. BCI Competition IV dataset 2a*

This dataset [33] contains EEG data acquired from 9 subjects. Several trials associated to four motor imagery tasks, namely the imagination of the left hand, right hand, both feet and tongue, were conducted. For each subject, two sessions of data were collected, with 288 trials per session (i.e., 72 trials for each class). The data is recorded from 22 channels, with sampling frequency of 250 Hz per channel and band pass filtered between 0.5 Hz and 100 Hz. The paradigm of the experiment is displayed in figure 2. At the start of the trial (t = 0 s), a fixation cross appeared on the screen along with a short acoustic tone. A cue in the form of an arrow pointing left, right, up, or down corresponding to left hand, right hand, tongue or foot MI task appeared on the screen at 2 s and remained there up to 3.25 s. The subject performed the MI task from 3 s to 6 s, after which a rest period is given for a duration of 1.5 s.

### 2.3. Pre-processing

Preprocessing of the data is carried out in order to reject noise and the undesirable physiological and non-physiological artifacts. Since the MI task patterns are prominently seen in the (8–30) Hz frequency band of the EEG, the data is band pass filtered in the range of (8–30) Hz. Due to the volume conduction, the raw EEG potentials measured at the scalp electrodes are known to have a poor spatial resolution. According to a simulation study in [34], sources within 6 cm radius of the recording electrode are having effect on the measured electrode potential. Spatial filters are used to suppress this spatial noise and extract the signal of interest. In this work, the Laplacian spatial filter is used which is explained in detail below.

*2.3.1. Laplacian spatial filter*

The Laplacian spatial filter acts as a spatial band pass filter, that emphasizes on activity originating in the radial sources below the electrode and reduces the impact of the widely distributed activity from distant sources [35]. Here, the signals from the adjacent four electrodes around a single electrode of interest (Ex: C3 or C4 for SMR signal) are averaged and this average is subtracted from the activity at the individual electrode [36]. The formula of the Laplacian filter is given by,

$$x_i^{LAP}(t) = x_i(t) - \sum\nolimits_{j\in S_i}(w_{ij}{}^*x_j(t)) \tag{1}$$

$$w_{ij} = \frac{1/d_{ij}}{\sum_{j\in S_i} 1/d_{ij}} \tag{2}$$

Here $x_i^{LAP}(t)$ is the spatial filtered signal of $i^{th}$ electrode, $x_i(t)$ is the potential of electrode $i$ with respect to reference electrode, $w_{ij}$ is the constant weight compute, $S_i$ is the set of electrodes surrounding the $i^{th}$ elctrode and $d_{ij}$ is the distance between electrodes $i$ and $j$, where $j$ is a member of $S_{i.}$

### 2.4. Feature extraction

The TF features are extracted from the pre-processed MI EEG data by using the complex Morlet Wavelet explained below.

*2.4.1. The complex morlet wavelet (CMW)*

This study uses a Complex Morlet Wavelet (CMW) as the mother wavelet which is generated from a complex sine wave windowed with a gaussian function. The equation of the CMW is given by:

$$CMW = \frac{1}{B_c\sqrt{2\pi}}\cdot e^{-\frac{1}{2}\left(\frac{t^2}{B_c{}^2}\right)}e^{j2\pi ft} \tag{3}$$

Here ‘t’ is the time, ‘f’ is the center frequency and $'B_c'$ is called as the time spread parameter or width of the gaussian function. The time spread parameter controls the time variance of the wavelet. Here $B_c = \text{n}/2\pi\text{f}$ and the parameter n is called as the number of wavelet cycles in the gaussian window. The value of n should be carefully chosen as it defines the trade-off between the temporal and frequency precision.

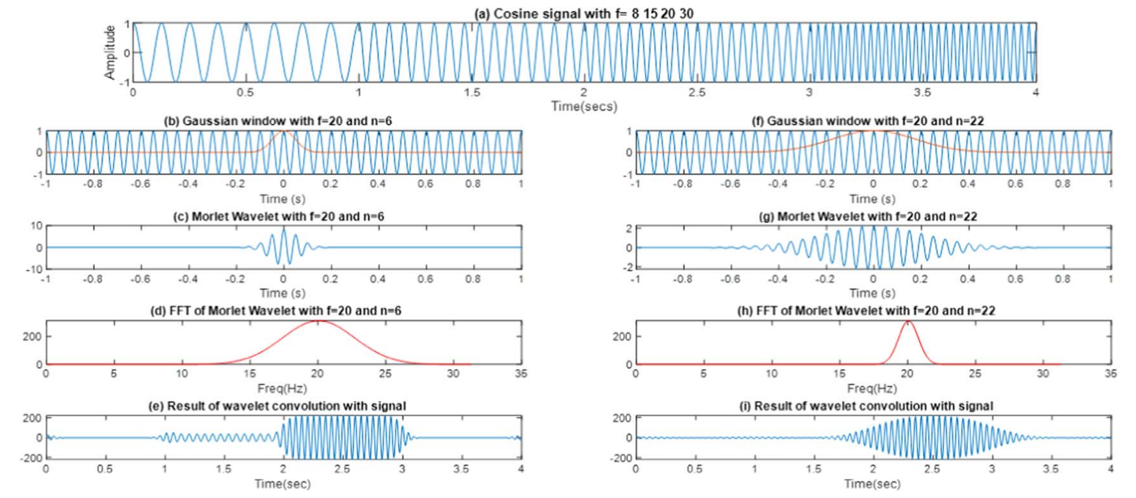


**Figure 3.** Analysis of a cosine signal by convolution with complex morlet wavelet.

*2.4.2. Impact of the value of n on resolutions*

Morlet wavelet with a higher value of n, is wider in time but has a narrower frequency spectrum. On the other hand, a lower value of n generates a Morlet wavelet that is narrower in time but has a wider frequency spectrum. In order to explore the impact of n on the time and frequency resolutions, a concatenated cosine signal with different frequencies is taken as the test signal for analysis. Figure 3(a) shows the concatenated cosine signal with a frequency of 8 Hz from 0 to 1 s, 15 Hz from 1 to 2 s, 20 Hz from 2 to 3 s and 30 Hz from 3 to 4 s. In the process of detecting the 20 Hz component present in the test signal, a cosine signal of frequency 20 Hz is overlapped with, a narrow gaussian window with $n = 6$ and a wider gaussian window with $n = 22$ as shown in figures 3(b) and (f) respectively, to generate Morlet wavelets of same frequency and different n values. The Morlet wavelet generated by multiplying the 20 Hz cosine signal with the gaussian window having $n = 6$ and $n = 22$ is shown in figures 3(c) and (g) respectively. The frequency spectrum of the Morlet wavelet with $n = 6$, shown in figure 3(d), is seen to be wider than the frequency spectrum of Morlet wavelet with $n = 22$, shown in figure 3(h). The test signal is convolved with the generated Morlet wavelets with different n values and their results are compared.

When n is equal to 6, which is much lower than the f value of 20 Hz, the convolution result gives a good time resolution as seen in figure 3(e), where the 20 Hz component is detected exactly in the 2 to 3 s time window. But it is also seen that a small magnitude of 15 Hz component is present in the 1 to 2 s time window thereby giving a poor frequency resolution. On the other hand, figure 3(i) shows that when n is chosen as 22, which is slightly nearer to the f value, the frequency resolution of the convolution result is very good as it has suppressed all the other frequencies other than 20 Hz. But a poor time resolution is seen here as there is an extension of the 20 Hz component before 2 s and after 3 s.

Hence, when the value of n is much lower than the measured frequency value, the frequency resolution of the convolution result undergoes degradation. In order to overcome the above shortcoming, the value of n is chosen around the frequency value. But a much higher n value causes a loss in the temporal resolution of the result. Hence choosing an optimum value for n is essential. In this work, the value of n is chosen relative to the frequency to be analyzed in order to generate the TFR with an optimum time and frequency resolution.

*2.4.3. Finding an optimal value of n*

To find the optimal value of n, a concatenated cosine signal is taken as the test signal for the analysis. Figure 4(a) shows a cosine signal with a frequency of 20 Hz from 0 to 1 s, 25 Hz from 1 to 2 s, 17 Hz from 2 to 3 s, 8 Hz from 3 to 4 s and 10 Hz from 4 to 5 s. Morlet wavelets with f values of 8 Hz, 10 Hz, 17 Hz, 20 Hz, and 25 Hz are generated for two different n values, n=6 and n=f, in order to compare their results. With the aim of detecting all the frequency components present in the test signal, all the generated wavelets are convolved with the test signal. The convolution results got by using the wavelets with n=6 are shown in figures 4(b), (c), (d), (e) and (f), while the convolution results got by using n=f are shown in figures 4(g), (h), (i), (j) and (k). For $n = 6$, it is observed that when the measured frequency range increases from 8 to 25 Hz, its frequency resolution degrades heavily. This is due to the fact that the value of n is much lower than the value of the measured frequency. On the flip side it is seen that when n=f, a very good frequency resolution is obtained for all the measured frequency components in the range of (8–25) Hz, but a considerable amount of loss in the temporal resolution is also observed here. In order to obtain the wavelet

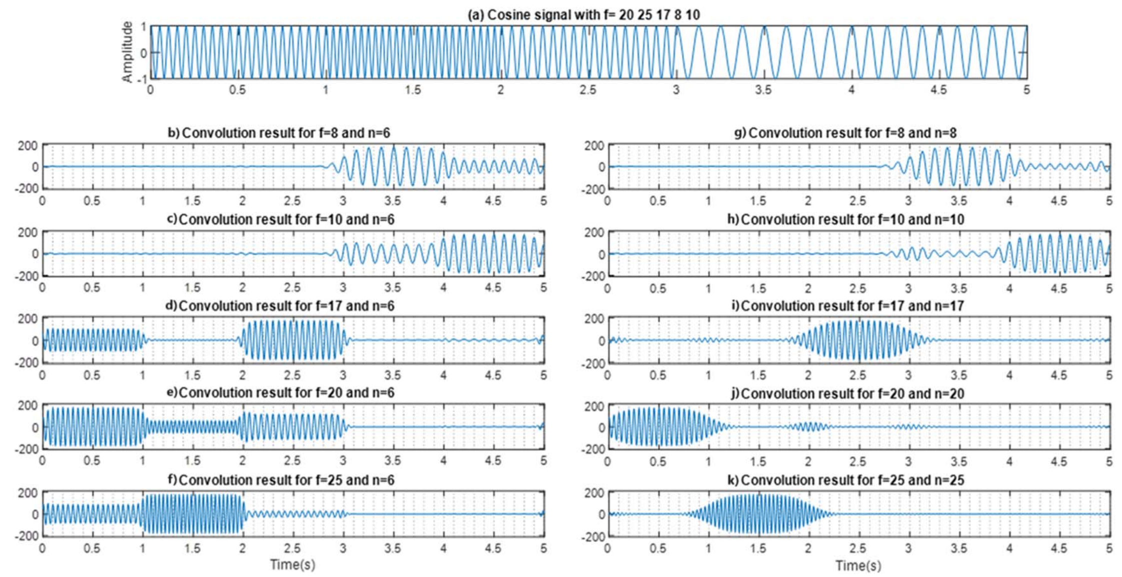


**Figure 4.** Analysis of convolution results at different values of n.

coefficients for each frequency, a heuristic set of 5 wavelets with a fixed central frequency and different values of n are generated as defined below:

$$CMW_f = \left\{\Psi_{f,n_i} | n_i = [f-4, f-2, f, f+2, f+4]\right\} \tag{4}$$

Here, $CMW_f$ is the Complex Morlet Wavelet set generated for a particular center frequency $f$. $\Psi_{f,n_i}$ is the CMW created for a center frequency $f$ and $n_i$ number of cycles. The range of values for $n_i$ is taken around the value of $f$ and is given by, $n_i$ = [f-4, f-2, f, f +2, f+4] where $i$ =1, 2….,5. The overall response of these wavelets for a particular frequency, $W_f$, is given by the Arithmetic Mean of the responses of the individual wavelets in the set.

$$W_f = \frac{\sum_{i=1}^{5} R\left[\Psi_{f,n_i}\right]}{5} \tag{5}$$

Here $R[\Psi_{f,n_i}]$ is the response of the $i^{th}$ wavelet in the set. It is obtained by convolving the resultant wavelet with the EEG signal $x$, as given below:

$$R\left[\Psi_{f,n_i}\right] = x^*\left[\Psi_{f,n_i}\right] \tag{6}$$

### 2.5. Formation of time frequency representations

In the dataset taken for this study, MI task cue is generated at 2 s and the MI task is performed by the subjects within the time window of 2 to 6 s. This study aims to effectively extract the MI task features using only 2 channels. Therefore, for each trial, the EEG data from the C3 and C4 channels is taken for a time duration of 2 to 6 s for the analysis. Since the sampling frequency of the data is 250 Hz, 1000 samples are obtained from each of the C3 and C4 channels for 144 left and 144 right hand trials of each subject. Figure 5 shows the process of formation of the TFR for a right hand MI trial data of subject 1 using a CMW. As the dataset uses 22 channels to collect data, the spatial noise present in C3 and C4 channels is suppressed by applying a Laplacian spatial filter. Figures 5(a) and 5(d) show the spatially filtered signal of the C3 and C4 channel respectively, that are got by subtracting the average of the signals present in the 4 channels surrounding the corresponding C3 or C4 channel.

The data is further band pass (8–30) Hz filtered since the MI task features are predominantly seen in this range and are shown in figures 5(b) and (e) for channels C3 and C4 respectively. The filtered signal is then convolved with the CMW generated for 20 values of center frequencies in the range of (8–14) Hz and (18–30) Hz as explained in section 2.4.3, in order to extract the time and frequency domain features from the preprocessed MI EEG signals. Complex valued wavelet coefficients for each center frequency are generated for the C3 and C4 channel data as a result of the convolution operations. The magnitude of these complex coefficients are calculated and further squared in order to find the power of the convolution results. These squared coefficient values of the C3 and C4 channels are stored in two separate matrices, as shown in figures 5(c) and (f) respectively, with a dimension of 20 × 1000 each. Here 20 rows represent the number of center frequencies used for analysis and 1000 represents the number of time samples. Further both these matrices are concatenated vertically to form a single matrix of dimension 40 × 1000, as shown in figure 5(g). This matrix forms the TFR of the MI EEG signal using the data from C3 and C4 channels and is generated for each of the 144 left and 144 right hand trials of the MI task data for each subject. As there are 9 subjects in the dataset, a total of 288 ×9 i.e., 2592 TFR matrices are generated. The features extracted by these TFR matrices are able to provide the energy of the MI EEG signals at individual frequencies and at different time

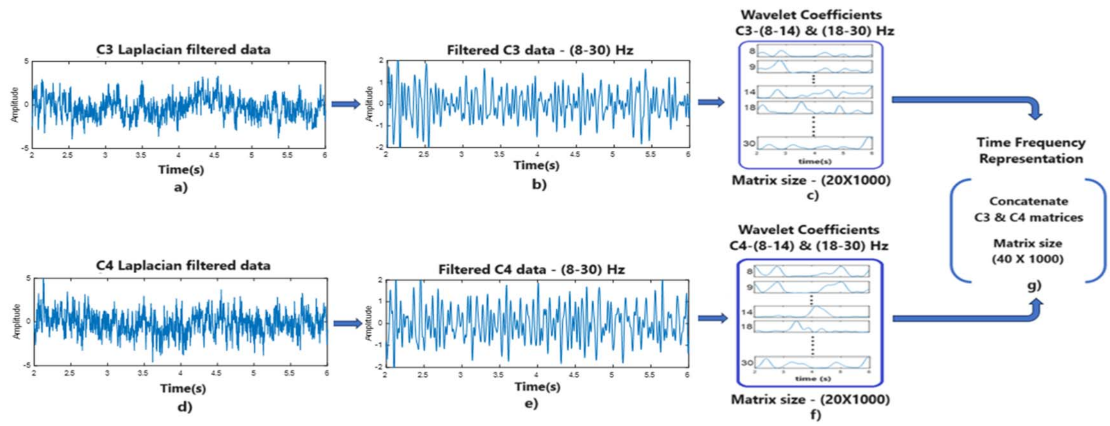


**Figure 5.** Formation of the TFR for a right hand MI trial of subject 1 from BCI Competition IV dataset 2a using CMW.

**Table 1.** Data split up strategy for training and testing the model.

| Subject | Total number of trials | | Total no of eliminated trials | | Used trials | | Training set | | Test set | |
|---|---|---|---|---|---|---|---|---|---|---|
| | LH | RH | LH | RH | LH | RH | LH | RH | LH | RH |
| A01 | 144 | 144 | 3 | 3 | 141 | 141 | 100 | 100 | 41 | 41 |
| A02 | 144 | 144 | 5 | 3 | 139 | 141 | 100 | 100 | 39 | 39 |
| A03 | 144 | 144 | 3 | 4 | 141 | 140 | 100 | 100 | 40 | 40 |
| A04 | 144 | 144 | 10 | 5 | 134 | 139 | 100 | 100 | 34 | 34 |
| A05 | 144 | 144 | 9 | 6 | 135 | 138 | 100 | 100 | 35 | 35 |
| A06 | 144 | 144 | 16 | 15 | 128 | 129 | 100 | 100 | 28 | 28 |
| A07 | 144 | 144 | 5 | 6 | 139 | 138 | 100 | 100 | 38 | 38 |
| A08 | 144 | 144 | 6 | 6 | 138 | 138 | 100 | 100 | 38 | 38 |
| A09 | 144 | 144 | 19 | 9 | 125 | 135 | 100 | 100 | 25 | 25 |

stamps with a higher resolution, as for the cosine signal discussed in section 2.4.3. Therefore, it provides a better representation of the ERD and ERS features during an MI task and will also be able to effectively extract the other time-frequency domain features from the MI data. These TFR matrices are further given as input to the CNN model for classification.

### 2.6. Data split up strategy for training and testing

The BCI competition 4 dataset 2a contains 144 left and 144 right hand trial data of 9 subjects. Some of the trials in the dataset are mentioned as rejected. Such trials are discarded for this study, and the CNN model is trained and tested using the remaining trials. Table 1 shows the number of rejected trials and the number of trials used for training and testing the CNN model for each of the 9 subjects. The TFR matrices of 200 trials, with 100 trials each for left and right hand, are randomly selected as the training set for each subject. The remaining TFR matrices are used for testing the model. From the training set, 20 TFRs of each class are randomly selected as the validation set, which is used for validation of the CNN model during the training process.

## 3. Convolutional neural network

### 3.1. Network architecture

The architecture of the CNN model built in this study is shown in figure 6. This network takes the TFR matrices generated for the MI data of 9 subjects as the input. The model is trained and tested for each of the subjects. Initially the TFR matrix of size (40 × 1000) for each trial of the MI task is given to the input layer of the CNN model. Further the data is passed on to the first convolutional layer that uses 8 small 2D kernels or filters of size (4 × 4), that are convolved with the input matrix with a stride of 1, thereby extracting informative features by fusing both the time and frequency domain information from the input data. The operations in the convolutional layers are represented by:

$$s = (x^{*}w) \tag{7}$$

Here $x$ is the input TFR matrix, $w$ is the kernel and $s$ is the resultant feature map for each kernel. Before training, the kernel values are initialized here as a random normal initialization. Convolution of the data with each of the 8 kernels results in 8 different feature maps each of size 37 ×997. The rectified linear unit i.e., ReLU activation function, shown in equation (8),

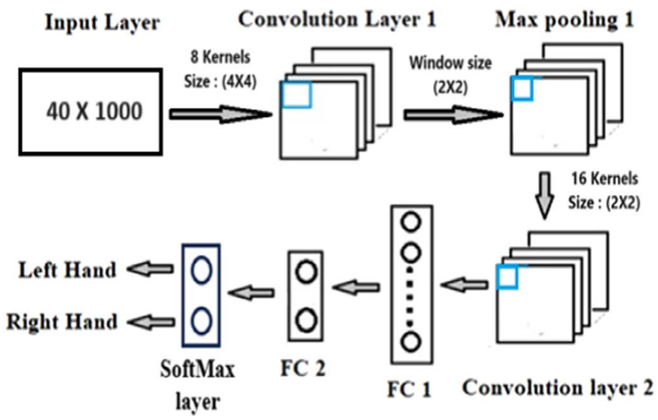


**Figure 6.** Architecture of the CNN model.

is applied to these feature maps, which returns zero for negative inputs and returns the same value for positive inputs. Hence, it transforms all the feature map values to a positive range.

$$Relu(x) = \max(0, x) \tag{8}$$

The 8 feature maps are now given to a max pooling layer having a window size of (2 × 2) and stride 1, that passes only the maximum value present in the windows, thereby highlighting the strong features and diminishing the weaker ones. The max pooling operation resizes each of the 8 feature maps to 36 × 996. After max pooling, a second convolutional layer with 16 randomly initialized kernels of size (2 × 2), having a ReLU activation function is placed. Each kernel operates on the 8 feature maps from the previous layer, extracting 8 different feature matrices, which are further summed up to generate a single feature map of dimension 35×995. Hence, 16 kernels generate 16 such feature maps each of size 35 × 995. The feature maps from the second convolutional layer are flattened to form a one-dimensional array f, of size 35 × 995 × 16 i.e., 557200 × 1 and are further passed on to the first fully connected (FC 1) layer with 10 neurons. Here each of the 10 neurons is connected to all the flattened 557200 neurons of the previous layer, thereby generating feature vector $Z_{FC1}$ of size (10 × 1). The output from FC 1 layer is then connected to the 2 neurons present in FC 2 layer thereby forming the global features of size 2 × 1 for classification.

This feature vector $Z_{FC2} = [z_1 z_2]^T$, is finally given to the SoftMax layer, that normalizes the real output feature values got from the FC2 layer to a probability value in the range (0,1). The final probability value $O = [o_1 o_2]^T$ predicted by the model for each class is computed by the SoftMax function as shown below:

$$o_1 = \frac{e^{z_1}}{e^{z_1} + e^{z_2}}; o_2 = \frac{e^{z_2}}{e^{z_1} + e^{z_2}} \tag{9}$$

### 3.2. Training and Hyperparameter settings

For every trial of the data, the probability values predicted by the model and the actual target labels given by $Y = [y_1 \; y_2]$, are used to compute the Cross Entropy Loss function, as shown below.

$$Lossfunction(L) = -[y_1 \ln(o_1) + y_2 \ln(o_2)] \tag{10}$$

Every trial can have a target label of [1 0], for the left hand MI task or [0 1], for the right hand MI task.

$$For \;\; Left \;\; hand \;\; trial: L = -y_1 \ln(o_1) \tag{11}$$

$$For \;\; Right \;\; hand \;\; trial: L = -y_2 \ln(o_2) \tag{12}$$

As updating the weights of the network for every trial of the data is computationally expensive and time consuming, a mini-batch of 32 trials are randomly selected from the training set of the MI task data. The loss for each of the 32 trials are computed and their average is taken in order to compute the aggregate loss function or error of the network, represented by $L_a$.This error value $L_a$ is back propagated into the network in order to update the weights by using the back propagation algorithm. When the entire training data flows through the model once, it is said to complete one epoch of training. The general equation for updating the weights of the model is given below:

$$W^{ij}_{new(MXN)} = W^{ij}_{old(MXN)} - \alpha \frac{\partial L}{\partial W^{ij}} \tag{13}$$

Here, $W^{ij}_{new}$ is the updated weight matrix present between the $i^{th}$ and $j^{th}$ layer of the model with a dimension of [MxN], $W^{ij}_{old}$ is the weight matrix to be updated, $\alpha$ is the learning rate parameter and $\frac{\partial L}{\partial W^{ij}}$ is the gradient of the loss function that gives the direction and the rate of change of the loss function with respect to the weights present between the $i^{th}$ and $j^{th}$ layers. Table 2 describes about all the layers used in the network and gives information about the dimension of the input and output matrices for each layer of the network. In order to update the weights $W^{89}_{(10X2)}$ present between the $8^{th}$ layer (FC1) and the $9^{th}$ layer

**Table 2.** Dimensions of the layers of the CNN model.

| No | Layer | Input dimension and operation | Output dimension |
|---|---|---|---|
| 1 | Input | $40 \times 1000 \times 1$ | $40 \times 1000 \times 1$ |
| 2 | Convolution 1 | $40 \times 1000 \times 1$, 8 filters $(4 \times 4)$, Stride 1 | $37 \times 997 \times 8$ |
| 3 | ReLU 1 | $37 \times 997 \times 8$ | $37 \times 997 \times 8$ |
| 4 | Max Pooling 1 | $37 \times 997 \times 8$, $(2 \times 2)$ Max pooling with Stride 1 | $36 \times 996 \times 8$ |
| 5 | Convolution 2 | $36 \times 996 \times 8$, 16 filters $(2 \times 2)$, Stride 1 | $35 \times 995 \times 16$ |
| 6 | ReLU 2 | $35 \times 995 \times 16$ | $35 \times 995 \times 16$ |
| 7 | Flatten layer | $35 \times 995 \times 16$ | $557200 \times 1$ |
| 8 | Fully connected layer 1 with 10 units | $557200 \times 1$ | $10 \times 1$ |
| 9 | Fully connected layer 2 with 2 units | $10 \times 1$ | $2 \times 1$ |
| 10 | Softmax | $2 \times 1$ | $2 \times 1$ |

(FC2) of the network, the derivative of the error function with respect to $W^{89}$ is calculated using the chain rule as shown below:

$$\frac{\partial L}{\partial W^{89}} = \frac{\partial L}{\partial O}\frac{\partial O}{\partial Z_{FC2}}\frac{\partial Z_{FC2}}{\partial W^{89}} = (O - Y)\cdot Z_{FC1} \tag{14}$$

The average of the above gradient is calculated for 32 trials and is used to update the weights as shown below:

$$W^{89}_{new(10\times 2)} = W^{89}_{old(10\times 2)} - \alpha\frac{\partial L}{\partial W^{89}} \tag{15}$$

The gradient is further calculated for the previous layer as shown below:

$$\frac{\partial L}{\partial W^{78}} = \frac{\partial L}{\partial Z_{\mathrm{FC2}}}\frac{\partial Z_{\mathrm{FC2}}}{\partial Z_{\mathrm{FC1}}}\frac{\partial Z_{\mathrm{FC1}}}{\partial W^{78}} = (O - Y)\cdot W^{89}\cdot f \tag{16}$$

$$W^{78}_{new(557200\times 10)} = W^{78}_{old(557200\times 10)} - \alpha\frac{\partial L}{\partial W^{78}} \tag{17}$$

It is observed that the error calculated in the output layer is weighted and passed back to the previous layers in the network. This process is continued until the initial layer is reached. Hence back propagation adjusts each weight of the network, proportionate to its contribution to the overall error $L_a$.

The network is trained using the data in the training set with a learning rate of 0.0001. During training, the performance of the model on the validation set is evaluated periodically after each epoch and the validation loss is calculated by taking the average of the losses obtained for each trial in the validation set. An early stopping strategy is used here for training the network in order to prevent overfitting. The network is made to stop training when validation loss becomes larger than or equal to the previous smallest loss for 15 epochs. The network converged for all the subjects within 40 epochs, when the loss function $L_a$ reached the minimum value. The data is shuffled while training the network after every epoch. The use of ReLU activation function avoids the vanishing gradient problem during back propagation of the error. The Adams optimization algorithm with a value of 0.9 for the gradient decay factor ($\beta_1$) and 0.99 for the squared gradient decay factor ($\beta_2$) is used here in order to have an adaptive learning rate for each parameter.

**Table 3.** Classification accuracy (%) of the CNN model for all subjects.

| Subject | Classification accuracy (%) | |
|---|---|---|
| | n value around f | n = f |
| A01 | 84.15 | 80.49 |
| A02 | 67.95 | 64.10 |
| A03 | 97.5 | 98.72 |
| A04 | 80.36 | 76.47 |
| A05 | 84.29 | 84.29 |
| A06 | 67.86 | 60.71 |
| A07 | 81.5 | 80.26 |
| A08 | 80.26 | 77.63 |
| A09 | 96.00 | 96.00 |
| Average | 82.20 | 79.85 |

## 4. Results and discussion

### 4.1. Comparison of results for different n values

This section gives the results of the proposed CNN model when trained and tested for each subject. Table 3 displays the classification accuracy of the model for the test data of each subject. The TFR matrices generated for n value equal to f and for n value around f are given as input to the CNN model and their classification results are compared. It is seen from the results that the average classification accuracy of the nine subjects when n value is around the f value is 82.20% and is higher than the 79.85% accuracy obtained for n = f. When n has a value around f, the subjects A01, A02, A04, A06, A07 and A08 give a higher accuracy value than n = f. But some subjects like A09 and A05 have the same accuracy for both n values. The highest classification accuracy is obtained by subject 3 and the lowest accuracy is obtained by subject 6. Hence the proposed method for choosing the value of n gives a better time and frequency representation of the MI data.

**Table 4.** Performance comparison with the existing methods for BCI Competition 4 dataset 2a.

| METHODS | ACCURACY (%) |
|---|---|
| CMW with CNN (Proposed) | 82.20 |
| CSP [37] | 76.23 |
| FBCSP [37] | 69.83 |
| CSP + LDA [38] | 69.75 |
| FBCSP + LDA [38] | 81.16 |
| ConvNets [38] | 79.90 |
| Filter bank and 3D CNN model [38] | 86.96 |
| Anchored STFT and Skip-net [28] | 81.2 |
| SW-LCR [39] | 80.0 |
| SW-Mode[39] | 79.8 |

### 4.2. Comparison with the existing methods

In table 4, the classification results obtained by the proposed method by taking the value of the number of cycles 'n' of the CMW around the value of centre frequency 'f', is compared with the accuracies obtained by the previous works using baseline methods like CSP, FBCSP, ConvNets and other existing methods, for the left and right hand MI task classification of dataset 2a from BCI competition 4. In [37], CSP and FBCSP have been applied on the dataset thereby giving accuracies of 76.23% and 69.83% respectively, which is much lower than accuracy obtained by the proposed method. Similarly, in [38] when CSP and FBCSP are used for feature extraction and LDA is used as the classifier, accuracies of 69.75% and 81.16% are respectively obtained. The proposed method has obtained an accuracy slightly higher than the FBCSP and LDA approach but performs much better than the CSP and LDA approach. An accuracy of 79.90% is obtained in [38] by using ConvNets, which is lower than the accuracy attained in this work. The method proposed in [38] uses the data from 22 channels to generate 3D spatiospectral feature representations of the MI data and further classifies them by using a 3D CNN model giving a classification accuracy of 86.96%, which is higher than the accuracies obtained by the proposed work and the baseline models. But the proposed model has outperformed the baseline methods by generating a TFR of the MI data from 2 channels, thereby having lesser complexity than the model used in [38]. The method proposed in [28] has classified the left and right hand MI tasks with an accuracy of 81.2% by using the MI EEG data from 3 channels, an anchored STFT for feature extraction and a skip-net CNN model with data augmentation techniques. The method proposed in this study has attained a slightly higher accuracy as compared to [28] by using the data from lesser number of channels and a CNN model with a simple architecture.

Similarly, it gives a higher accuracy than [39] which uses Sliding Window (SW) to extract nine timeframes of length 1 sec from the MI data of multiple channels and uses CSP and LDA for feature extraction and classification of data in each frame respectively. Hence the proposed method performed significantly better than the baseline and existing methods for the left and right hand MI task classification.

## 5. Conclusion and future work

The CMW has been used in this work for extracting the high-resolution TFRs of the MI tasks. A new method for the selection of the value of the number of cycles for the CMW relative to the center frequency is analyzed here. It gives a better representation of the energy of the signals at individual frequencies and at different time stamps, thereby influencing the classification performance of the CNN. The proposed framework has used data from two channels i.e., C3 and C4, of the BCI competition 4 dataset 2a for generating the TFR matrices and a CNN model with simple architecture is used for classification. The TFR matrices generated for the MI tasks with n value around the f value are found to have a higher average classification accuracy of 82.20% as compared to the TFR matrices generated with n = f. The proposed model has obtained a higher classification accuracy than the baseline methods and some of the existing methods for left and right hand MI task classification using data from lesser number of channels. However, the CNN model can be further improvised for a better performance. The CNN model used here is subject dependent and hence requires more time for training the model for individual subjects. Hence for the future work, subject independent CNN models can be developed which saves time and allows new users to quickly access the BCI systems, thereby extending the proposed framework to real-time applications. Further, visualizing the features learned by the layers of the CNN can provide a clear interpretation of the decisions made by the CNN model. The novel method used to define the value of the number of cycles of the CMW has been applied only on the MI BCI dataset. In future this can be applied to the other EEG based BCI modalities or to the different research areas of EEG signals in order to check their efficiency in the extraction of the time and frequency domain features across domains.

## Data availability statement

No new data were created or analysed in this study.

## Author contributions

All authors contributed to the study conception and design.

V Srimadumathi: Conceptualization, Methodology, Software, Investigation, Formal analysis, and writing original draft was done. Ramasubba Reddy M:

Conceptualization, Validation, Resources, Writing-Review & Editing, Supervision was done.

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## Conflict of interest

The authors have no relevant financial or non-financial interests to disclose.

## Ethical approval

This study did not require ethics approval.

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